EnergyMind Tunisia – *Mizan*

Anomaly Detection for a Trustworthy Smart-Metered Grid

Phase 1 technical note: types of anomalies, families of detectors, what we use in *Mizan*, and why.
All figures are generated from code (data files + TikZ/PGFPlots).

In one paragraph. Our system must decide, for every suspicious reading, whether the *grid* did something unusual or the *data* is wrong, because the two need opposite responses. No single detector can do this. We therefore combine (i) physics and statistical rules, which are auditable, (ii) three unsupervised machine-learning detectors trained only on clean data, which are sensitive to subtle patterns, and (iii) a cross-sensor consistency reasoner that separates genuine events from data-quality problems. Section 5 backs this design with a small, honestly-labelled experiment.

1 The Problem: a Suspicious Number Has Two Possible Meanings

An anomaly is a pattern that deviates from expected behaviour [1]. In our feeder (one transformer, about 50 smart meters, rooftop PV) a surprising value can come from the **grid** or from the **data chain**, and the right response differs: a real event needs operational action, a faulty or falsified sensor needs to be distrusted and repaired. A further difficulty is that an informed attacker can craft false data that passes standard bad-data checks [3], so one test is never enough.

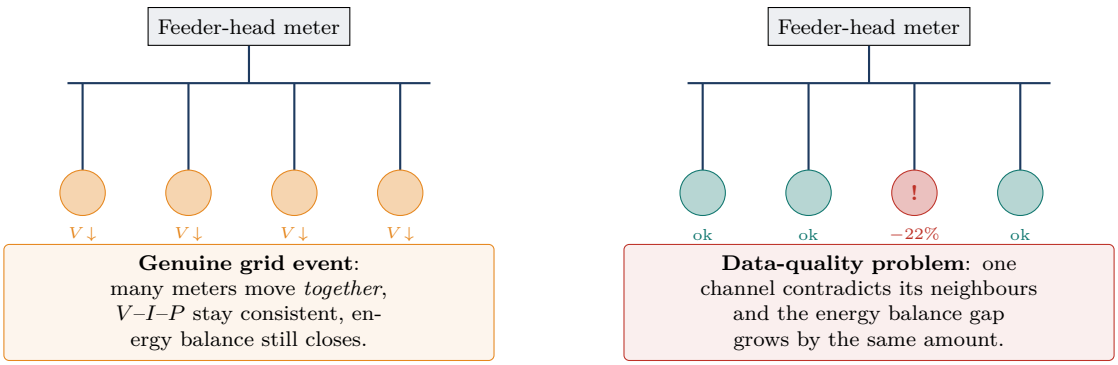


Figure 1: The core question. Left: a real event is visible on many sensors and obeys physics. Right: a faulty or falsified channel disagrees with redundant measurements. This cross-sensor logic is the backbone of our design.

2 Types of Anomalies

The literature classifies anomalies by how they relate to context [1]. We use that classification and map it onto grid failures (Figure 2, Table 1).

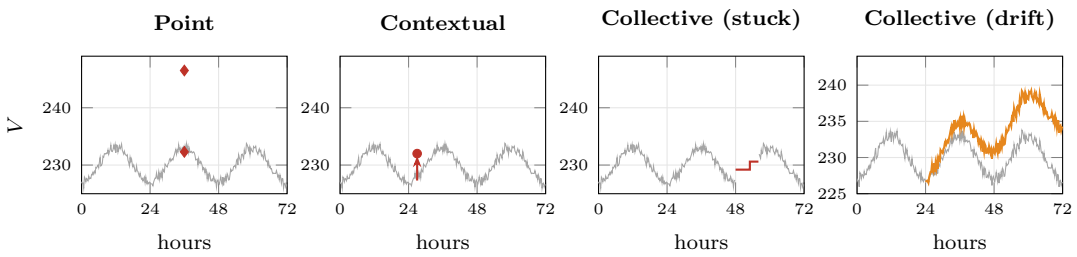


Figure 2: Illustrative voltage traces (synthetic, 15-min samples). Grey: normal behaviour. Red/amber: the anomaly. A *point* anomaly is one odd value; a *contextual* anomaly is a value that is normal in general but wrong for its time of day; *collective* anomalies are odd only as a sequence (a frozen sensor, a slow drift).

Project anomaly	**Type [1]**	**Signature**	**First line of defence**
Spike, dropout	Point	Isolated impossible value	Range/rate limits, robust z
Stuck sensor	Collective	Variance collapses to ≈ 0	Flat-line rule, autoencoder
Slow drift, gain error	Collective	Gradual offset from expected profile	CUSUM, residual vs. neighbours
Constant bias	Contextual	Plausible but systematically shifted	Energy balance, OCSVM
Peak-hour theft	Contextual	Reduction only in specific hours	Time-of-day model, CUSUM
Replay / stealth injection	Collective	Looks normal per channel	Cross-sensor consistency [3]
Genuine surge / outage	(not an error)	Many channels, physics holds	Spatial-extent check

Table 1: From anomaly types to our project taxonomy. The last row must *not* be flagged as bad data.

3 Families of Detection Methods and Our Choice

Methods can be organised by *what knowledge they use* and by *how many labels they need* [1, 2]. Because labelled faults are rare in grids, we are driven toward the semi-supervised and unsupervised regimes: learn what "normal" looks like from clean data, and flag deviations (Figure 3).

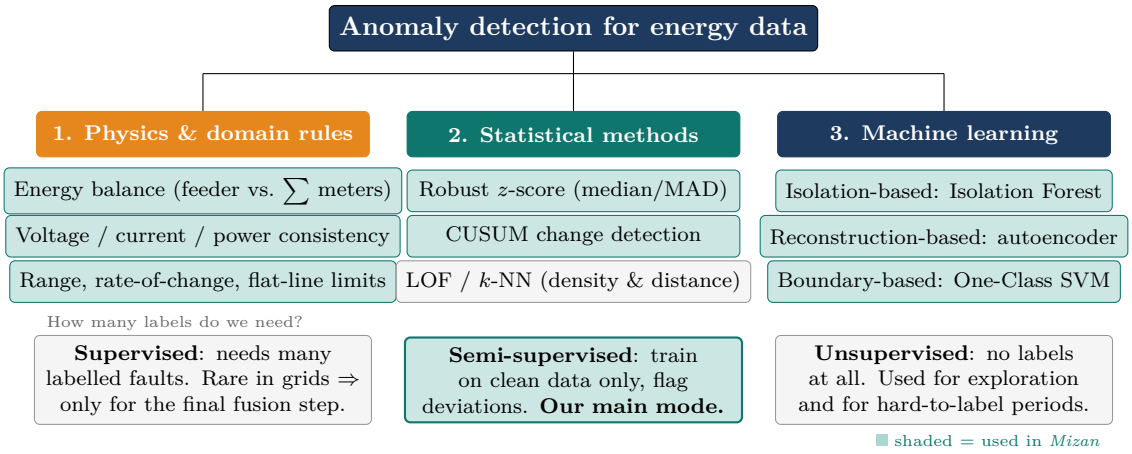


Figure 3: Taxonomy of detectors. Teal boxes are the methods we use (detailed in Section 4). Hybrid physics + learning follows the argument that high-stakes decisions should rest on inspectable models where possible [4].

4 The Methods, One by One

4.1 Statistical Outlier Methods

Robust z-score. The classic $z = (x - \bar{x})/s$ is distorted by the very outliers it seeks, because mean and standard deviation have a 0% breakdown point. Replacing them by the median and the scaled median absolute deviation gives

$$z_i^{\text{rob}} = \frac{x_i - \text{med}(x)}{1.4826 \cdot \text{MAD}(x)}, \qquad \text{MAD} = \text{med}_i |x_i - \text{med}(x)|, \tag{1}$$

which tolerates up to 50% contamination [5, 6]. **CUSUM** [7] accumulates small deviations, $S_t = \max(0,\, S_{t-1} + (x_t - \mu_0)/\sigma - k)$, and alarms when $S_t > h$; it is the natural tool for slow drift. **LOF** [8] compares a point's local density with that of its neighbours, so it handles clusters of different density.

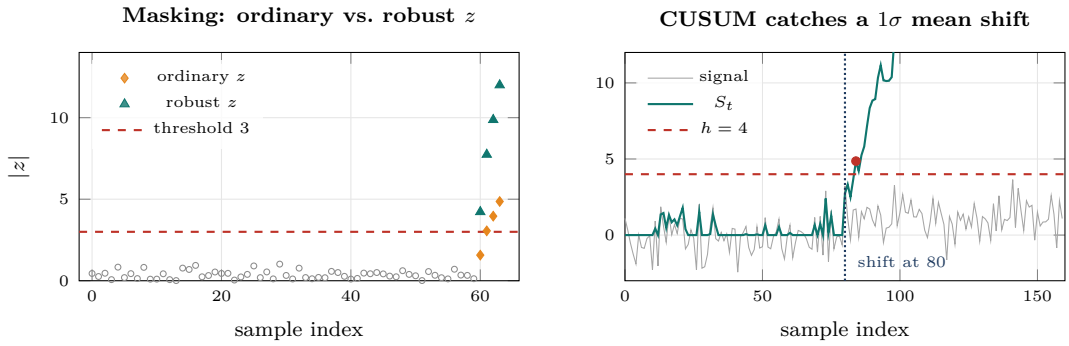


Figure 4: Left: four injected outliers among 60 normal samples. Ordinary z gives 1.6, 3.1, 4.0, 4.9, so 1 of them hides below the threshold (masking); robust z gives 4.2, 7.7, 9.9, 12.0, so all four are flagged. Right: CUSUM ($k = 0.5$, $h = 4$) alarms at sample 84, 4 samples after the shift, although the raw signal stays mostly within $\pm 3\sigma$. Both from our `gen.py`, synthetic data, fixed seed.

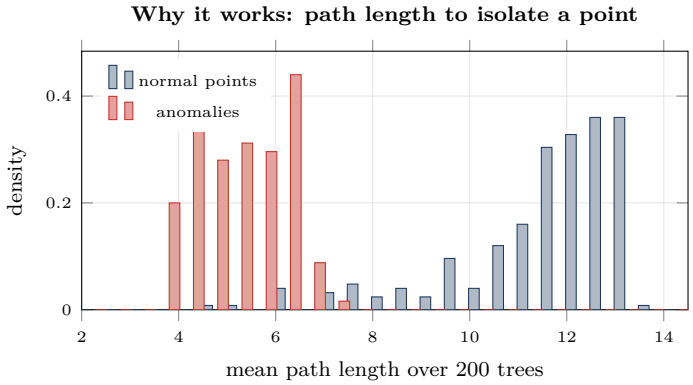
Role in *Mizan*. Cheap, transparent first line and the baseline every ML component must beat. Applied to residuals from a time-of-day profile, so the evening peak is not mistaken for an anomaly (contextual anomalies).

4.2 Isolation Forest

Liu, Ting and Zhou [9] observed that anomalies are few and different, so random axis-parallel cuts isolate them in fewer splits. A tree is grown by repeatedly picking a random feature and a random split value; the path length $h(x)$ to isolate x is short for anomalies. With subsample size ψ the score is

$$s(x,\psi) = 2^{-E[h(x)]/c(\psi)}, \qquad c(\psi) = 2H(\psi-1) - \frac{2(\psi-1)}{\psi}, \qquad H(i) \approx \ln i + 0.5772, \tag{2}$$

so $s \to 1$ for anomalies and $s < 0.5$ for normal points. It is linear in time, needs no labels and no distance computations. The Extended Isolation Forest uses randomly oriented cuts and removes the axis-parallel artefacts visible in score maps [10], which matters for correlated quantities such as V and I.



Reading the plot. Normal points sit deep inside dense regions, so many random cuts are needed to separate them (mean ≈ 11.6). Anomalies are isolated after only 5.2 cuts on average. A short average path means a high score s. Same kind of 2-D cloud as Figure 6.

Figure 5: Average isolation path length from the fitted Isolation Forest (200 trees; 250 normal points and 250 anomalies drawn outside the dense core).

4.3 One-Class SVM

Schölkopf et al. [11] separate the data from the origin in a kernel feature space with maximum margin:

$$\min_{w,\xi,\rho} \ \tfrac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho \quad \text{s.t.} \quad \langle w, \phi(x_i)\rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \tag{3}$$

and $f(x) = \operatorname{sgn}(\langle w, \phi(x)\rangle - \rho)$ with an RBF kernel. The parameter ν upper-bounds the fraction of training points treated as outliers. The closely related SVDD [12] encloses the data in a minimal hypersphere.

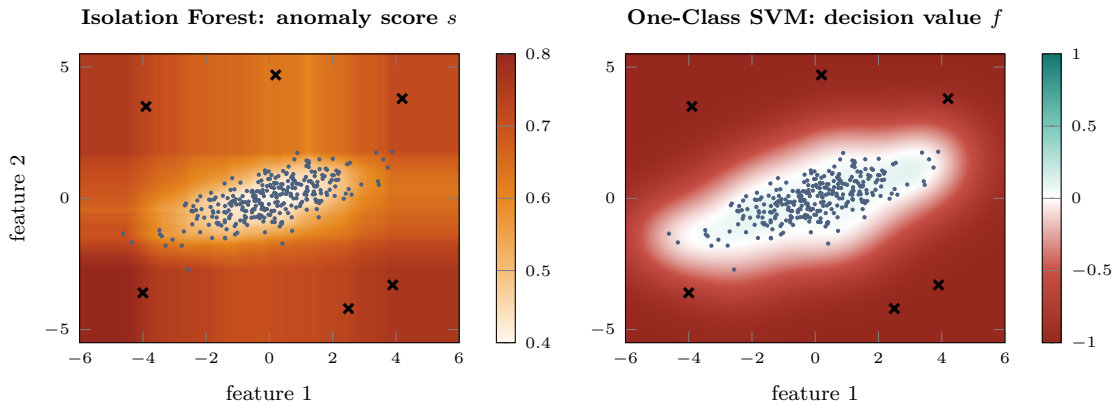


Figure 6: Two detectors fitted by `scikit-learn` [16] on the same correlated 2-D "normal" cloud (blue dots); crosses are injected outliers. Isolation Forest (200 trees, $\psi = 256$) scores the outliers 0.63–0.77 against a mean of 0.46 for normal points. The One-Class SVM ($\nu = 0.05$, RBF; decision value scaled to $[-1, 1]$) draws a smooth boundary (white) around the data. Their different inductive biases are why disagreement carries information.

Role in *Mizan*. Both run on window features of each meter (mean, spread, max residual, slope, autocorrelation, share of zero increments). Isolation Forest is our fast first ML pass. OCSVM is trained on a subsample (it scales poorly) as a second, differently-biased opinion. When they disagree, confidence drops and the case is routed to a human.

4.4 Autoencoders

An autoencoder learns to compress and reconstruct normal data through a narrow code. Trained on clean data only, it reconstructs anomalies badly, so the reconstruction error is the anomaly score [13]:

$$e(x) = \|x - g(f(x))\|^2 = \textstyle\sum_j e_j, \qquad e_j = (x_j - \hat{x}_j)^2. \tag{4}$$

LSTM encoder–decoders extend this to multi-sensor time series [14]; surveys of deep detectors discuss when they help and where they over-generalise and reconstruct anomalies too well [2]. Crucially for us, the *per-channel* error e_j points at *which sensor* looks wrong, and it can be fed to Shapley-value explanations [15].

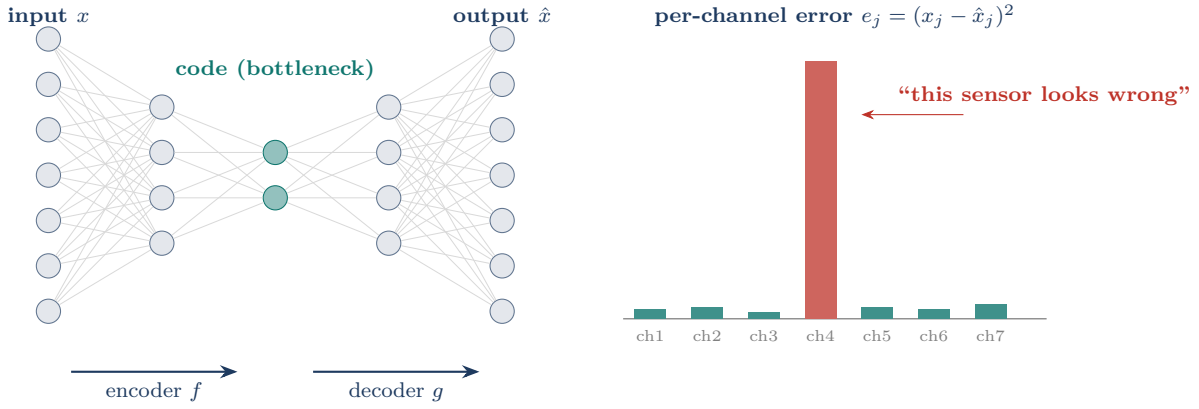


Figure 7: Autoencoder principle (left) and the per-channel reconstruction error that localises the faulty sensor (right; illustrative values).

Role in *Mizan*. Trained on clean windows only; captures correlated-sensor inconsistencies and subtle waveform changes that thresholds miss, and gives channel-level evidence for the

explanation sentence ("channel 4 deviates most").

4.5 The Methods at a Glance

Method	Labels	Run cost	Explainable	Strong on	Weak on
Rules / physics	none	very low	by design	Impossible values, balance gaps	Needs correct physics model
Robust z / CUSUM	none	very low	high	Spikes; small persistent shifts	Multivariate patterns
Isolation Forest	none	low	medium (TreeSHAP)	Global outliers, many features	Shifts inside the normal range
One-Class SVM	none	high on big data	low (post-hoc)	Smooth boundary on compact data	Scale; kernel tuning
Autoencoder	none	medium (training)	medium (per channel)	Correlated-sensor, waveform change	Can reconstruct anomalies too well

Table 2: Qualitative comparison, our assessment from the cited literature [1, 2] and the experiment in Section 5.1. Green = favourable, amber = acceptable, red = costly.

5 Putting It Together: Our Hybrid Detection Stack

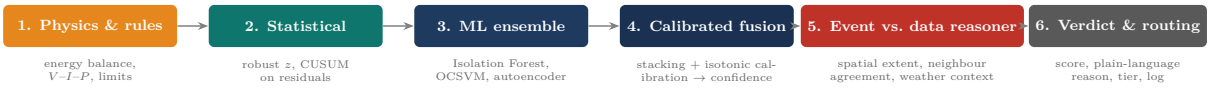


Figure 8: *Mizan*'s detection pipeline. Rules can veto or force escalation; ML scores add sensitivity; the reasoner decides *what kind* of unusual it is.

How the verdict is reached. Layers 1–3 answer "is something unusual?"; layer 5 answers "what kind?" from consistency features (Table 3).

Evidence	Genuine event	Data-quality problem
Spatial extent (meters affected)	Many	One or a few
Physical consistency (V, I, P, balance)	Holds	Violated on one channel
Agreement with correlated neighbours	Yes	Contradicts them
Temporal shape	Physical dynamics	Step, frozen, replayed
Weather / time-of-day context	Explains it	Does not explain it

Table 3: Consistency features used by the event-versus-data reasoner.

5.1 A Small, Honest Experiment

To show why one detector is not enough we ran a toy experiment (all code in `gen.py`). We simulate 12-hour windows of one meter channel (48 samples at 15 min, evening-peak load profile, 6% noise). The five detectors are fitted on 3 000 clean windows; each threshold is set for a 1% false-positive rate on 1 500 validation windows; 400 injected windows per anomaly type are then scored on fresh data.

Detector (% recall)	Spike	Stuck	Drift	Bias	Noise	Theft	Replay	FPR
Robust z (max)	78	64	7	11	78	22	2	2.1%
CUSUM	38	84	36	59	62	64	1	1.1%
Isolation Forest	62	84	15	16	86	53	2	1.5%
One-Class SVM	66	88	30	66	86	62	2	1.4%
Autoencoder	54	75	10	30	91	50	2	0.9%
Ensemble (any of 5)	76	85	30	61	88	63	3	2.5%

Table 4: Recall (%) per anomaly type on held-out windows. Darker = better. Injection ranges: spike ±18–42% of one sample; stuck 12–23 samples; drift linear +2–6%; bias ×1.015–1.05; noise $\sigma = 0.07$; theft −12% at the evening peak; replay = a normal window from another day. The last row fires if any detector fires, each set at 0.25% FPR. FPR is measured on 1 500 clean test windows. **Toy, single-channel, simulated; it illustrates behaviour, it is not a performance claim for the final system.**

What the experiment shows.

- **No detector wins everywhere.** Robust z is best on spikes (78%) but nearly blind to drift (7%) and bias (11%); CUSUM is best on drift (36%) and peak-hour theft (64%); OCSVM on stuck sensors (88%) and bias (66%); the autoencoder on added noise (91%).

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